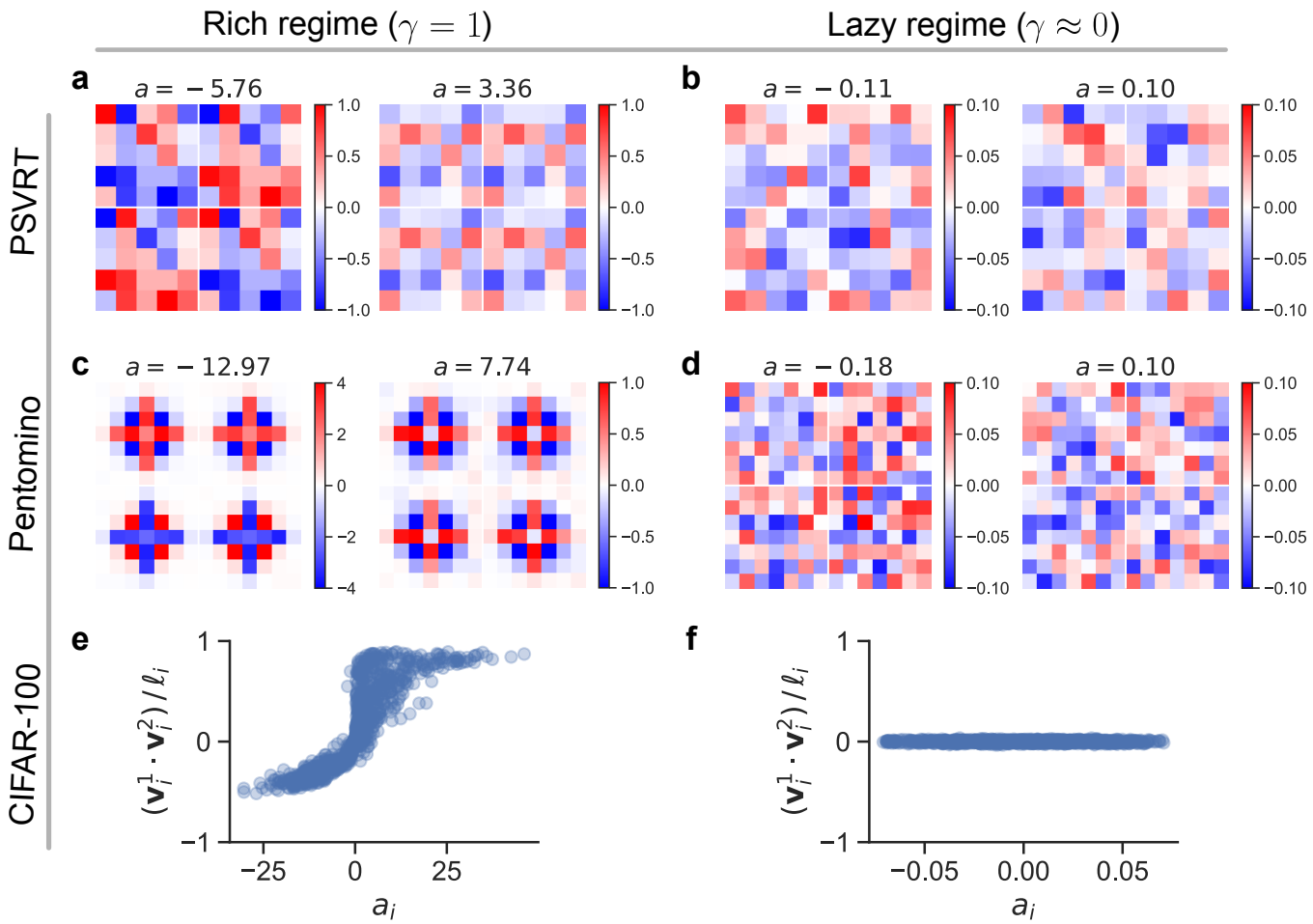

Figure A1: **Conceptual representations in visual same-different tasks. (a)** Visualization of representative hidden weights associated with maximal positive and negative readout weights, for a rich model training on PSVRT. Parallel/antiparallel structures are visible in how regions are either the same or precisely the opposite of their neighbors. **(b)** Visualization of representative hidden weights associated with maximal positive and negative readouts, for a lazy model training on PSVRT. There is no discernible structure, and the magnitudes of both readouts and hidden weights are small. **(c)** The same as (a), but for a rich model on the Pentomino task. Parallel/antiparallel structures are likewise visible. **(d)** The same as (b), but for a lazy model on the Pentomino task. There is no discernible structure, and the magnitudes of both readouts and hidden weights are small. **(e)** For a rich model trained on the CIFAR-100 task, we plot the alignment between weight components corresponding to the two images. There is a distinct parallel/antiparallel-like structure visible in the weight alignment, as we saw for MLPs trained on our simple SD task. **(f)** For a lazy model trained on the CIFAR-100 task, we plot the alignment between weight components corresponding to the two images. There are no discernible parallel/antiparallel structures at all.

which is certainly a positive quantity. Hence, the model classifies all positive examples correctly.

Upon receiving a *different* example $\mathbf{x}^- = (\mathbf{z}, \mathbf{z}')$, our model returns

$$f(\mathbf{x}^-) = \phi(\mathbf{1}\cdot\mathbf{z} + \mathbf{1}\cdot\mathbf{z}') + \phi(-\mathbf{1}\cdot\mathbf{z} - \mathbf{1}\cdot\mathbf{z}') - \rho(\phi(\mathbf{1}\cdot\mathbf{z} - \mathbf{1}\cdot\mathbf{z}') + \phi(-\mathbf{1}\cdot\mathbf{z} + \mathbf{1}\cdot\mathbf{z}'))$$

Since training symbols are sampled as $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I}/d)$, we have that $\mathbf{z} \overset{d}{=} -\mathbf{z}$. Furthermore, a sum of independent Gaussians remains Gaussian, so $\mathbf{1}\cdot\mathbf{z} \pm 1\cdot\mathbf{z}' \sim \mathcal{N}(0, 2)$. Hence, $f(\mathbf{x}^-) \overset{d}{=} u - \rho v$, where $u, v \sim \text{HalfNormal}(0, 2)$. Note, the ReLU nonlinearity ensures these quantities are distributed along a Half-Normal distribution, rather than a Gaussian. Further, u and v are independent since $\mathbf{1}\cdot\mathbf{z} + \mathbf{1}\cdot\mathbf{z}'$ is independent from $\mathbf{1}\cdot\mathbf{z} - \mathbf{1}\cdot\mathbf{z}'$ (the two sums are jointly Gaussian with zero covariance).

The test accuracy of the model on $\mathbf{x}^-$ is given by $p(f(\mathbf{x}^-) < 0)$, which can be expressed as an integral over the joint PDF of u, v:

$$p(f(\mathbf{x}^-) < 0) = p(u - \rho v < 0) = \frac{1}{2\pi}\int_0^\infty \int_{u/\rho}^\infty \exp\left\{-\frac{u^2+v^2}{8}\right\} dv\,du\,.$$

To compute this quantity, we convert to polar coordinates. Let $u = r\cos(\theta)$ and $v = r\sin(\theta)$. Under this change of variables, we have

$$\begin{aligned} p(f(\mathbf{x}^-) < 0) &= \frac{1}{2\pi}\int_{\tan^{-1}(1/\rho)}^{\pi/2}\int_0^\infty r\exp e^{-r^2/8}\,dr\,d\theta \\ &= \frac{2}{\pi}\left(\frac{\pi}{2} - \tan^{-1}(1/\rho)\right) \\ &= \frac{2}{\pi}\tan^{-1}(\rho)\,. \end{aligned}$$

For $\rho \to \infty$, this quantity approaches 1. Since ρ cancels in the result for the *same* input, ρ can be arbitrarily large without impacting the classification accuracy on *same* inputs. Hence, the hand-crafted solution overall solves the same-different task provided the relative magnitude of the negative readouts is large.

A technical detail required for both successful positive and negative classifications is that the test example is not precisely orthogonal to the parallel/antiparallel vectors, in which case the relevant dot products would be zero. However, a test example is exactly orthogonal to the weight vectors with probability zero, so this eventuality does not impact the solution's overall test accuracy.

Figure B1 illustrates how parallel/antiparallel weight vectors may correctly classify a *same* or *different* example.

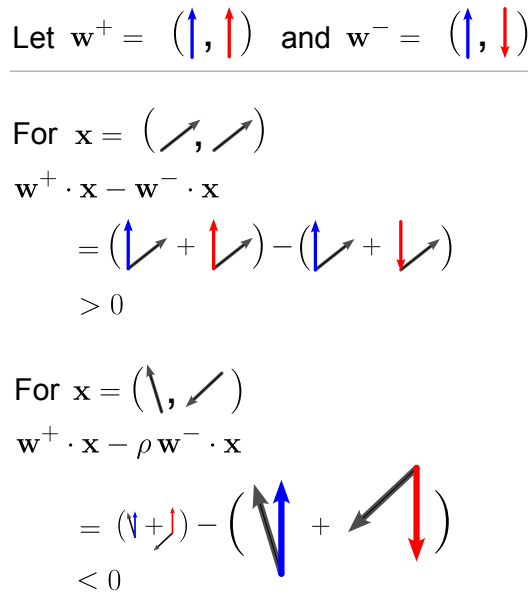

Figure B1: **Illustration of hand-crafted solution.** Parallel/antiparallel weight vectors $\mathbf{w}^+, \mathbf{w}^-$ are represented pictorially as sets of two vectors. The dot product operation is represented by conjoining the corresponding vectors: the dot product equals the cosine angle scaled by the magnitudes of the component vectors. For a *same* test example, the $\mathbf{w}^+ \cdot \mathbf{x}$ remains positive while $\mathbf{w}^- \cdot \mathbf{x}$ cancels to zero. For a *different* test example, the relative magnitude ρ enables a successful negative classification.

C Rich regime details

We conduct our analysis of the rich regime in two parts. We begin with a derivation of the max margin solution to our same-different task in Section C.1. Doing so requires us to replace our model's ReLU activations with quadratic activations. The max margin solution also does not demonstrate the rich model's learning efficiency or insensitivity to perceptual details. To address these shortcomings, we extend our analysis by considering a heuristic construction in which we approximate a rich MLP using an ensemble of independent Markov processes (Section C.2). Using this construction, we derive a finer-grain characterization of the MLP's weight structure, and apply it to estimate the model's test accuracy for varying L and d.

C.1 Max margin solution

An MLP trained on a classification objective often learns a max margin solution over the dataset (Morwani et al., 2023; Chizat & Bach, 2020; Wei et al., 2019). While this outcome is not guaranteed in our setting, studying the structure of the max margin solution nonetheless reveals critical details about how our MLP may be solving the same-different task. Following Morwani et al. (2023), we adopt two conditions to expedite our analysis:

1. We replace a strict max margin objective with a max *average* margin objective over a dataset $\mathcal{D} = \{\mathbf{x}_n, y_n\}_{n=1}^P$

$$\max_\theta \frac{1}{P}\sum_{n=1}^P \left[(2y_n - 1) f(\mathbf{x}_n; \theta)\right],$$

where $\mathbf{x}_n, y_n$ are sampled over a training distribution with L symbols and the objective is subject to some norm constraint on θ. Given the symmetry of the task, a max average margin objective forms a reasonable proxy to the strict max margin.

2. We consider quadratic activations $\phi(\cdot) = (\cdot)^2$ rather than ReLU. Doing so alters our model from Eq (1), but we later use a heuristic construction to argue that the resulting solution is recovered under ReLU activations in a rich learning regime.

We further allow $P, L \to \infty$. Following these simplifications, we derive the max average margin solution.

Theorem 1. *Let $\mathcal{D} = \{\mathbf{x}_n, y_n\}_{n=1}^P$ be a training set consisting of P points sampled across L training symbols, as specified in Section 2. Let f be the MLP given by Eq 1, with two changes:*

1. *Fix the readouts $a_i = \pm 1$, where exactly $m/2$ readouts are positive and the remaining are negative.*
2. *Use quadratic activations $\phi(\cdot) = (\cdot)^2$.*

For weights $\theta = \{\mathbf{w}_i\}_{i=1}^m$, define the max margin set $\Delta(\theta)$ to be

$$\Delta(\theta) = \arg\max_{\theta} \frac{1}{P} \sum_{n=1}^{P} \left[(2y_n - 1) f(\mathbf{x}_n; \theta)\right],$$

subject to the norm constraints $||\mathbf{w}_i|| = 1$. If $P, L \to \infty$, then for any $\mathbf{w}_i = (\mathbf{v}_i^1; \mathbf{v}_i^2) \in \Delta(\theta)$ and $\ell_i = \left|\left|\mathbf{v}_i^1\right|\right| \left|\left|\mathbf{v}_i^2\right|\right|$, we have that $\mathbf{v}_i^1 \cdot \mathbf{v}_i^2 / \ell_i = 1$ if $a_i = 1$ and $\mathbf{v}_i^1 \cdot \mathbf{v}_i^2 / \ell_i = -1$ if $a_i = -1$. Further, $\left|\left|\mathbf{v}_i^1\right|\right| = \left|\left|\mathbf{v}_i^2\right|\right|$.

Proof. Let $\mathcal{D}^+$ be the subset of $\mathcal{D}$ consisting of *same* examples and $\mathcal{D}^-$ be the subset of *different* examples. Let I^+ be the set of indices i such that the readout weight $a_i > 0$. Let I^- be the set of indices j such that $a_j < 0$. Then our max average margin solution becomes

$$\begin{aligned}
\max_{\theta} \frac{1}{P} \sum_{n=1}^{P} \left[(2y_n - 1) f(\mathbf{x}_n; \theta)\right] &= \max_{\theta} \frac{1}{P} \left[\sum_{\mathbf{x}^+ \in \mathcal{D}^+} \left[f(\mathbf{x}^+; \theta)\right] - \sum_{\mathbf{x}^- \in \mathcal{D}^-} \left[f(\mathbf{x}^-; \theta)\right] \right] \\
&= \max_{\theta} \frac{1}{P} \left[\sum_{\mathcal{D}^+} \left[\sum_{i \in I^+} |a_i| (\mathbf{w}_i \cdot \mathbf{x}^+)^2 \right] - \sum_{\mathcal{D}^-} \left[\sum_{i \in I^+} |a_i| (\mathbf{w}_i \cdot \mathbf{x}^-)^2 \right] \right] \\
&\quad + \frac{1}{P} \left[\sum_{\mathcal{D}^-} \left[\sum_{j \in I^-} |a_j| (\mathbf{w}_j \cdot \mathbf{x}^-)^2 \right] - \sum_{\mathcal{D}^-} \left[\sum_{j \in I^-} |a_j| (\mathbf{w}_j \cdot \mathbf{x}^+)^2 \right] \right].
\end{aligned}$$

Suppose we stack all *same* training examples $\mathbf{x}^+$ into a large matrix $\mathbf{X}^+ \in \mathbb{R}^{|\mathcal{D}^+| \times 2d}$ and stack all *different* training examples $\mathbf{x}^-$ into a large matrix $\mathbf{X}^- \in \mathbb{R}^{|\mathcal{D}^-| \times 2d}$. Applying the norm constraints $||\mathbf{w}_i|| = 1$, our max margin solution is resolved by the following objectives

$$\begin{aligned}
\mathbf{w}_*^+ &= \arg\max_{\mathbf{w}} \frac{1}{P} \left[\left|\left|\mathbf{X}^+ \mathbf{w}\right|\right|^2 - \left|\left|\mathbf{X}^- \mathbf{w}\right|\right|^2 \right] \quad \text{such that } ||\mathbf{w}|| = 1, \\
\mathbf{w}_*^- &= \arg\min_{\mathbf{w}} \frac{1}{P} \left[\left|\left|\mathbf{X}^+ \mathbf{w}\right|\right|^2 - \left|\left|\mathbf{X}^- \mathbf{w}\right|\right|^2 \right] \quad \text{such that } ||\mathbf{w}|| = 1,
\end{aligned}$$

where $\mathbf{w}_*^+$ and $\mathbf{w}_*^-$ represent the hidden weights of the max average margin solution.

Maximizing (or minimizing) this objective is equivalent to finding the largest (or smallest) eigenvector of the matrix $\mathbf{X} = \frac{1}{P}\left[(\mathbf{X}^+)^\intercal \mathbf{X}^+ - (\mathbf{X}^-)^\intercal \mathbf{X}^-\right]$. In the limit $P, L \to \infty$, this matrix becomes circulant. Let us see how. Note that $\mathbf{X} \in \mathbb{R}^{2d \times 2d}$. Suppose there are exactly $P/2$ *same* examples and $P/2$ *different* examples. Along the diagonal of $\mathbf{X}$ are terms

$$X_{ii} = \frac{1}{P} \sum_{j=1}^{P/2} \left[(x_{ij}^+)^2 - (x_{ij}^-)^2 \right],$$

where x_{ij}^+ corresponds to the ith index of the jth *same* example, and x_{ij}^- is the same for the jth *different* example. Because $L \to \infty$, we have that $x_{i,j}^+ \sim \mathcal{N}(0, 1/d)$ and $x_{i,j}^- \sim \mathcal{N}(0, 1/d)$, where $x_{i,j}^+$ is independent of $x_{i,j}^-$. Hence, $X_{ii} \to 0$ as $P \to \infty$.

Now let us consider the diagonal of the first quadrant

$$X_{i,2i} = \frac{1}{P} \sum_{j=1}^{P/2} \left[x_{ij}^+ x_{2i,j}^+ - x_{ij}^- x_{2i,j}^- \right].$$

For same examples, $x_{ij}^+ = x_{2i,j}^+$, so

$$\frac{1}{P} \sum_{j=1}^{P/2} x_{ij}^+ x_{2i,j}^+ = \frac{1}{P} \sum_{j=1}^{P/2} \left(x_{ij}^+\right)^2 \to \frac{1}{2} \mathbb{E}\left[\left(x_{ij}^+\right)^2\right] = \frac{1}{2d}$$

For different examples, x_{ij}^- remains independent of $x_{2i,j}^-$, so

$$\frac{1}{P} \sum_{j=1}^{P/2} x_{ij}^+ x_{2i,j}^+ \to 0.$$

We therefore have overall that $X_{i,2i} \to 1/2d$. The same argument applies for the diagonal of the third quadrant, revealing that $X_{2i,i} \to 1/2d$.